import and display data

```
import pandas as pd
Df = pd.read_csv("/content/creditriskdataset.zip")
Df
```

ion	personal_status	other_parties	...	property_magnitude	cc_age	other_payment_p
4	male single	none	...	real estate	67	
2	female div/dep/mar	none	...	real estate	22	
2	male single	none	...	real estate	49	
2	male single	guarantor	...	life insurance	45	
3	male single	none	...	no known property	53	
...	...	...	...	...	...	
3	female div/dep/mar	none	...	real estate	31	
4	male div/sep	none	...	life insurance	40	
4	male single	none	...	car	38	
4	male single	none	...	no known property	23	
3	male single	none	...	car	27	

Value_count() It is a useful method which can get count of each category in a categorical attributed series of values

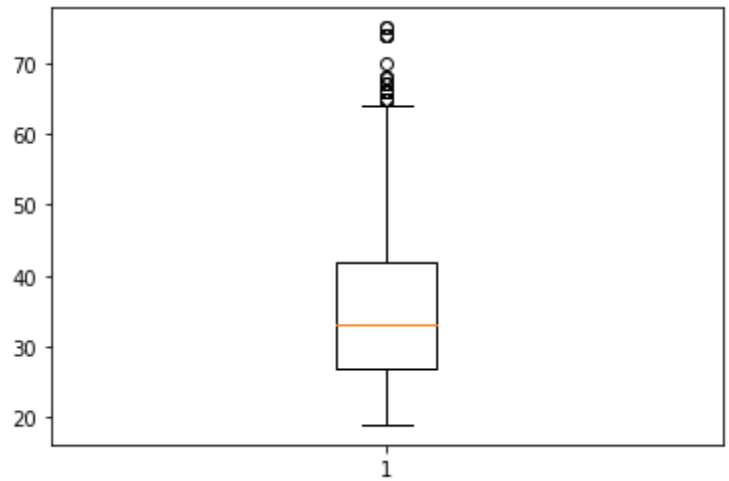
```
Df["credit_usage"].value_counts()
```

```
24    184
12    179
18    113
36     83
6      75
15     64
9      49
```

```
48     48
30     40
21     30
10     28
60     13
27     13
42     11
11      9
20      8
8       7
4       6
45      5
7       5
39      5
14      4
13      4
33      3
28      3
54      2
16      2
22      2
47      1
5       1
26      1
72      1
40      1
Name: credit_usage, dtype: int64
```

BoxPlot- It can be used to pictorically represent distribution of data which shows extreme values, median and quartiles. Outliers can be easily found using boxplots

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3 DF = pd.read_csv("/content/creditriskdataset.zip")
4 y = list(DF.cc_age)
5 plt.boxplot(y)
6 plt.show()
```

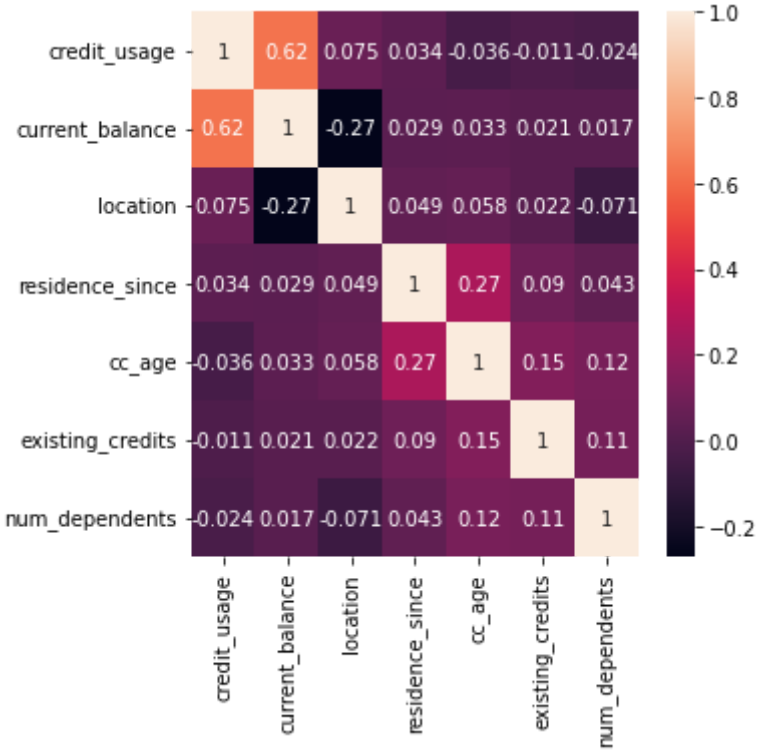



```
1 import numpy as np
2 from google.colab import drive
```

```
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
import seaborn as sns
```

```
correlation_matrix = Df.corr()
# annot = True to print the values inside the square
plt.subplots(figsize=(5,5))
sns.heatmap(data=correlation_matrix, annot=True)
```

```
<matplotlib.axes._subplots.AxesSubplot at 0x7f02de00dc90>
```



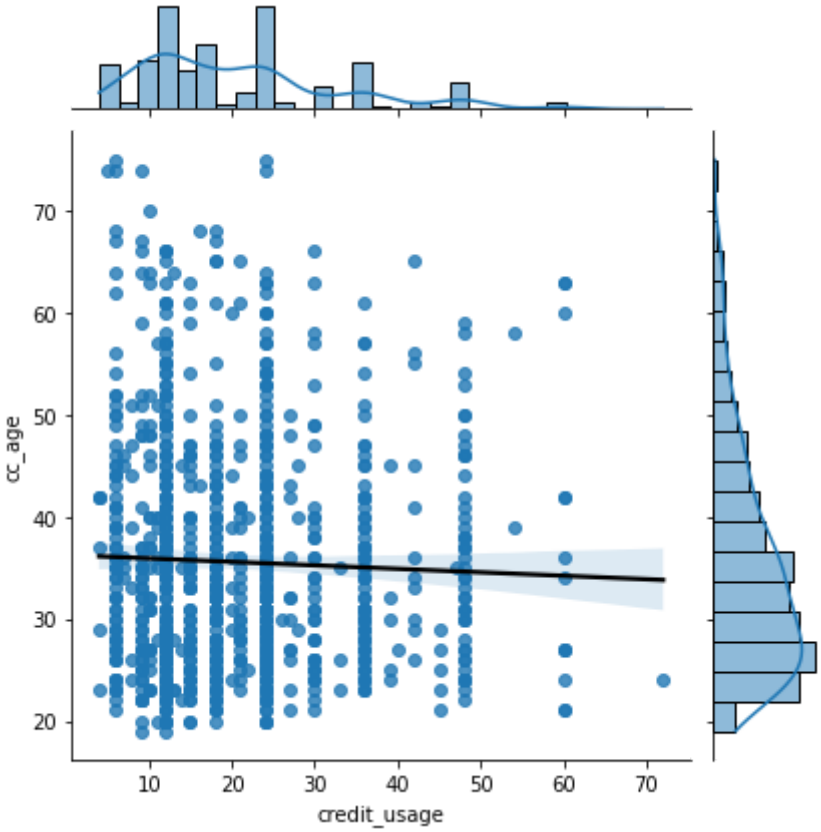
The above plot is a correlation matrix. A correlation matrix is used to visualize how the features are related to each other or with the target variable.

```
Df['credit_usage'].describe()
sns.distplot(Df['credit_usage']);
```

```
/usr/local/lib/python3.7/dist-packages/seaborn/distributions.py:2619: FutureWarning:
  warnings.warn(msg, FutureWarning)
```

Above is a histogram for credit usage. It can be seen that the usage is high between 10-20 on the x axis.

```
1  import seaborn as sns
2  g = sns.jointplot(x="credit_usage", y="cc_age", data=Df, kind='reg')
3  regline = g.ax_joint.get_lines()[0]
4  regline.set_color('black')
5  regline.set_zorder(3)
```



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